Traffic Light Control Using ATmega16

COE 332 | Microcontroller Programming

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Project Overview

This project simulates a three-color traffic signal using the ATmega16 microcontroller in Proteus. The red, yellow, and green lights turn on sequentially for 10, 3, and 10 seconds, respectively, and the sequence repeats continuously.

Hardware and Timing

Signal	ATmega16 Port	Physical Pin	ON Duration
Red	PB0	1	10 seconds
Yellow	PB1	2	3 seconds
Green	PB2	3	10 seconds

Clock frequency: 16 MHz (configure the Proteus microcontroller clock accordingly).

Source Code (main.c)

```
#define F_CPU 16000000UL
#include <avr/io.h>
#include <util/delay.h>

int main(void)
{
    DDRB = 0x07;  // PB0, PB1, PB2 as outputs

    while (1)
    {
        // RED LED - 10 seconds
        PORTB = 0x01;
        for (int i = 0; i < 10; i++)
        {
            _delay_ms(1000);
        }

        // YELLOW LED - 3 seconds
        PORTB = 0x02;
        for (int i = 0; i < 3; i++)
        {
            _delay_ms(1000);
        }

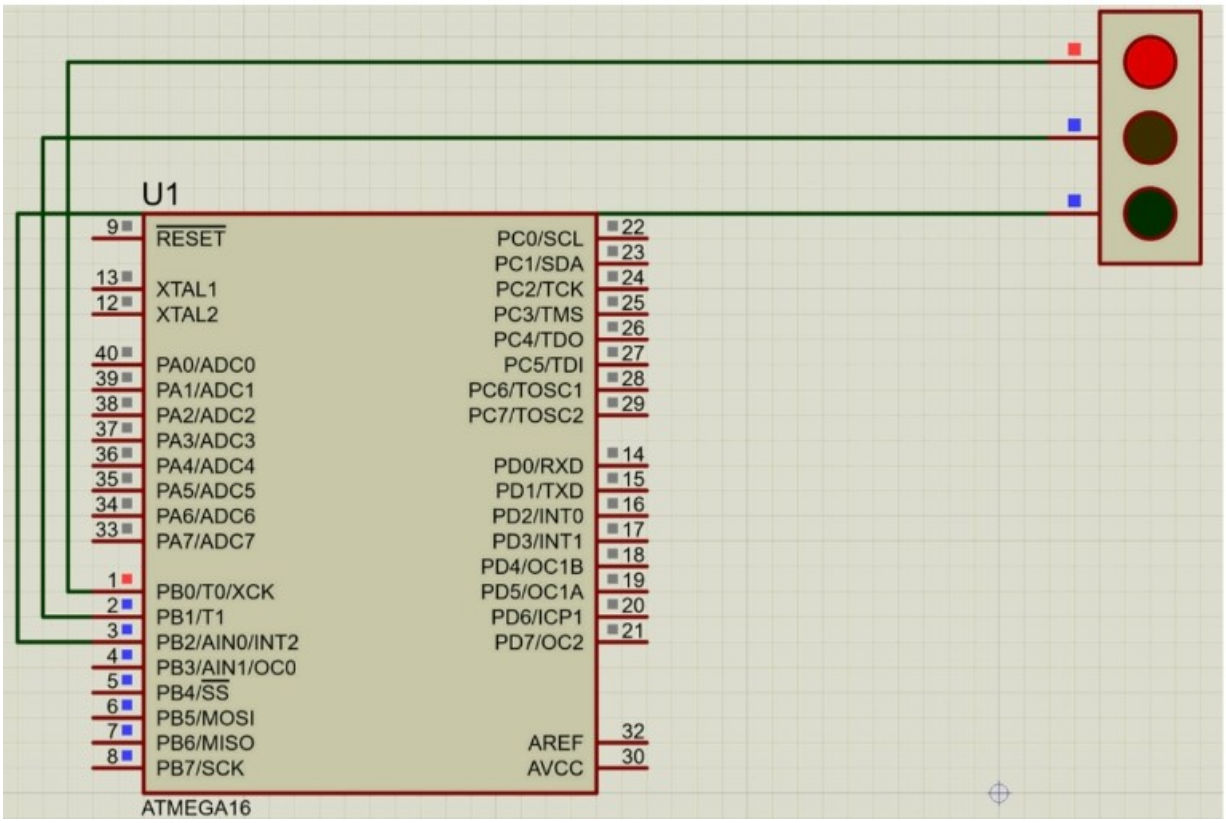
        // GREEN LED - 10 seconds
        PORTB = 0x04;
        for (int i = 0; i < 10; i++)
        {
            _delay_ms(1000);
        }
    }
}
```

Hardware Configuration

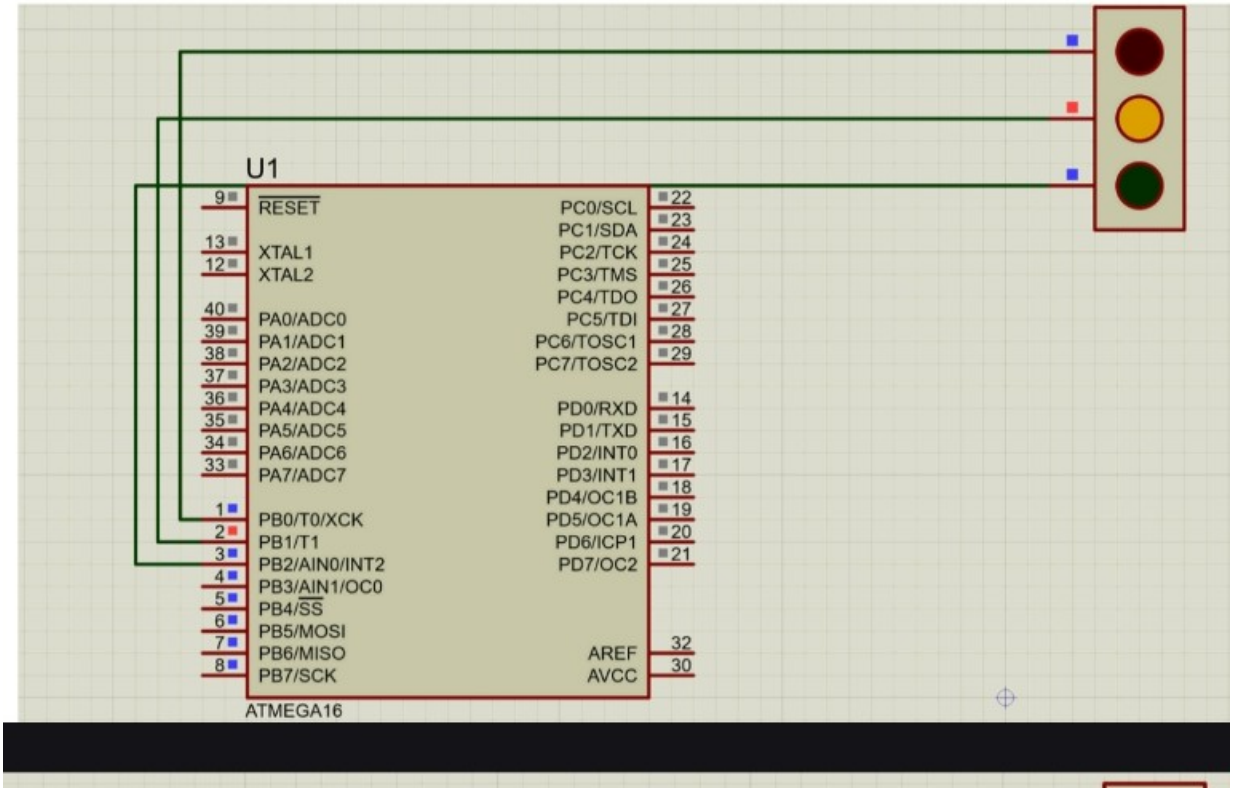
ATmega16 | PORTB: PB0 - Red (10 s), PB1 - Yellow (3 s), PB2 - Green (10 s).

Proteus Simulation Results

1. Red light ON



2. Yellow light ON



3. Green light ON

